John Duncan

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SUMMARY

Self-taught software engineer with substantial independent, well-tested project evidence across Python, Swift, and C#. Independently designed, built, and deployed a full-stack Python/FastAPI platform to Google Cloud Run on PostgreSQL, and built two native iOS applications in Swift and migrated both to Unity/C# — over 8,000 automated tests across the three, with compile-time enforced module boundaries and formal architecture decision records. Professional experience includes contract Python and AI-evaluation work and a C#/Unity game programming internship. Strengths: backend services and cloud deployment, iOS and Unity architecture, and automated testing.

PROFESSIONAL EXPERIENCE

Game Programmer (Internship)

[P1] Virtual Civilization Initiative — Remote, CA · Nov 2023 – Nov 2024

- Debugged and supported the team with code adjustments in C# using the Unity project manager.

Software Developer / AI Developer (Contract)

DataAnnotation — Remote, CA · Dec 2023 – Mar 2024

- Created a Python algorithm to sort and process text files to specification.
- Tested and debugged AI model responses for quality and correctness.

Education Specialist

Alexander Twilight College Preparatory Academy (Aspire Public Schools) — Sacramento, CA · Jul 2023 – Present

- Developed data tracking and progress-monitoring systems to improve intervention fidelity and support data-informed decision making.

SELECTED PROJECTS

- **SPEDstack — Educational Workflow Platform** — Independently developed and deployed a full-stack educational workflow platform using Python, FastAPI, SQLAlchemy, PostgreSQL, Docker, and Google Cloud Run. Built instructional planning and scheduling tools, backend services, and an automated test suite of 5,930 passing tests. Sole author of ~116,000 lines across 238 commits.
- **ReadingAvatar — K–3 Literacy Learning Application** — Designed and built a K–3 literacy application (Swift/iOS, migrated to Unity/C#) in which a student's reading aloud drives the gameplay across phonics, fluency, vocabulary, and comprehension. Implemented on-device speech recognition and a mastery estimator that weights evidence by how it was gathered, with design decisions recorded as architecture decision records; ~1,100 automated tests.
- **VisualAvatar — Individualized Visual Supports (iPad)** — Built a native iPad application delivering individualized visual supports — visual cues, first-then boards, choice boards, and task sequences — for students with disabilities. Architected it to run fully offline and on-device with no student accounts and no data transmission, enforced by an automated architecture check that fails the build if networking, analytics, or third-party dependencies enter shipping code. Swift and Unity/C# with module boundaries enforced at compile time; ~1,065 automated tests plus documented on-device verification.

TECHNICAL & PROFESSIONAL SKILLS

Software Engineering: Python (FastAPI, Pydantic, SQLAlchemy, pytest); Swift / SwiftUI (iOS, SPM, XCTest); C# / Unity (asmdef modules, EditMode/PlayMode tests); JavaScript / React (CRA, React Router, Bootstrap, GSAP); Cloud deployment: Google Cloud Run, Cloud SQL (Postgres), Docker; LLM/API integration (Anthropic, OpenAI); Google Workspace / Sheets API automation

Product Engineering: Accessibility & assistive-technology design (individualized visual supports, speech-driven interaction); Privacy-by-design for student data (fully on-device architecture, no accounts, no data transmission); Product scoping & requirements definition (documented scope and design decisions); Software architecture & architecture decision records (ADRs); Automated testing & release verification (pytest, XCTest, Unity Test Framework)

EDUCATION

M.A. in Education — Alder Graduate School of Education (Jun 2023)

B.S. in Kinesiology — Sacramento State University (2020)